University of Massachusetts, Amherst
CICS
COMPSCI 683: Artificial Intelligence

Assignment 5: Regret Minimization and Intro to Logic

Due: two weeks after the issue date in Gradescope

Important Instructions:

- Please upload your solution to Gradescope.
- You may work with a partner on your homework; however, if you choose to do so, **you must** write down the name of your partner on the assignment. In addition, you **may not submit similar/identical solutions**.
- Provide a mathematical proof/computation steps to all questions, unless explicitly told not to. Solutions with no proof/explanations (even correct ones!) will be awarded no points.
- You may look up combinatorial inequalities and mathematical formulas online, but please do not look up the solutions. Most problems can be found in research papers/online; if you happen upon the solution before figuring it out yourself, mention this in the submission. **Unreferenced copies of online material will be considered plagiarism.**

A knowledge base KB is a set of logical rules that model what the agent knows. These rules are written using a certain language (or *syntax*) and use a certain truth model (or *semantics* which say when a certain statement is true or false). In propositional logic sentences are defined as follows

1. Atomic Boolean variables are sentences.
2. If S is a sentence, then so is $\neg S$.
3. If S_1 and S_2 are sentences, then so is:
   (a) $S_1 \wedge S_2$ "S_1 and S_2"
   (b) $S_1 \vee S_2$ "S_1 or S_2"
   (c) $S_1 \Rightarrow S_2$ "S_1 implies S_2"
   (d) $S_1 \Leftrightarrow S_2$ "S_1 holds if and only if S_2 holds"

We say that a logical statement a models b ($a \models b$) if b holds whenever a holds. In other words, if $M(q)$ is the set of all value assignments to variables in a for which a holds true, then $M(a) \subseteq M(b)$.

An inference algorithm $\mathcal{A}$ is one that takes as input a knowledge base KB and a query α and decides whether α is derived from KB, written as $KB \vdash_{\mathcal{A}} \alpha$. $\mathcal{A}$ is sound if $KB \vdash_{\mathcal{A}} \alpha$ implies that $KB \models \alpha$; $\mathcal{A}$ is complete if $KB \models \alpha$ implies that $KB \vdash_{\mathcal{A}} \alpha$.

1. (30 points) Given the following logical statements, use truth-table enumeration to show that $KB \models \alpha$. In other words, write down all possible true/false assignments to the variables, the ones for which KB is true and the one for which α is true, and see whether one is a subset of the other.
   (a) (15 points)

$$KB = (x_1 \vee x_2) \wedge (x_1 \Rightarrow x_3) \wedge \neg x_2$$
$$\alpha = x_3 \vee x_2$$

(b) (15 points)

$$KB = (x_1 \vee x_3) \wedge (x_1 \Rightarrow \neg x_2)$$
$$\alpha = \neg x_2$$

2. (15 points) We have proven the following bound on the regret of the multiplicative weights update algorithm:

$$R_{\texttt{MWU}}(T) \leq \frac{\log n}{\varepsilon T} + \varepsilon$$

Suppose that we know T in advance, and that $T > \log n$. Derive the value of $\varepsilon \in [0, 1]$ that minimizes the regret. What is the regret bound obtained under this value of ε?

3. (25 points) Here are two sentences in the language of first-order logic:

(a) $\forall x : \exists y : (x \geq y)$

(b) $\exists y : \forall x : (x \geq y)$

(a) (7 points) Assume that the variables range over all the natural numbers $0, 1, 2, \ldots$ and that the "$\geq$" predicate means "is greater than or equal to". Under this interpretation, translate (a) and (b) into English.

(b) (8 points) Is (a) true under this interpretation? Is (b) true under this interpretation?

(c) (10 points) Does (a) logically entail (b)? Does (b) logically entail (a)? Justify your answers.

4. (30 points) Two English sentences "Anyone who takes an AI course is smart" and "Any course that teaches an AI topic is an AI course" have been represented in first-order logic:

$$\forall x : (\exists y : \texttt{AI\_Course}(y) \wedge \texttt{Takes}(x, y)) \Rightarrow \texttt{Smart}(x)$$

$$\forall x : (\exists y : \texttt{AI\_Topic}(y) \wedge \texttt{Teaches}(x, y)) \Rightarrow \texttt{AI\_Course}(x)$$

(a) (10 points) It is known that Yvonne takes COMPSCI 683 and COMPSCI 683 teaches Inference, which is an AI topic. Represent these facts as first-order logic sentences.

(b) (10 points) Next, convert all first-order logic sentences into conjunctive normal form.

(c) (10 points) Use resolution to prove the statement "Yvonne is smart."

Use the following table format to represent the steps you take. Here, sentences 1 and 2 are the ones you are resolving. $\theta =$ is the substitutions you are making to unify them (if any). The resolvent is (as its name suggests) the sentence that results from resolving the two sentences you picked. For brevity, you can use the number of the sentence instead of copying verbatim. For example, if my KB has two facts (1) and (4) as clauses I want to resolve, I can write (1) in sentence 1, (2) in sentence 2, and assign a new number to the resolvent (if I want to use it in the future).

Step #	Sentence 1	Sentence 2	Resolvent	$\theta =$

5. (Bonus points) In what follows, we will prove that 3-CNF formulas with a small number of clauses will always have a satisfying assignment. Recall that a 3-CNF formula ϕ comprises of clauses of 3 literals each. For example,

$$\phi(\vec{x}) = (x_1 \vee \neg x_2 \vee \neg x_3) \wedge (x_2 \vee \neg x_3 \vee \neg x_5) \wedge (x_2 \vee x_4 \vee \neg x_5) \wedge (x_1 \vee \neg x_4 \vee x_5)$$

is a 3-CNF formula with four clauses of size 3 each. We assume that *each clause has exactly three different variables in it*. For example, we will not have clauses of the form $(x_2 \vee x_2 \vee x_3)$ or $(x_7 \vee \neg x_8 \vee \neg x_7)$. Suppose that we choose the truth value of each varibable i.i.d. uniformly at random. In other words, we set

$$x_i = \begin{cases} \mathit{True} & \text{with probability } \frac{1}{2} \\ \mathit{False} & \text{otherwise.} \end{cases}$$

We call this a *uniform random assignment of variables*.

(a) (Bonus points) Consider a 3-CNF formula with m clauses and n variables. What is the probability that a clause C in ϕ is satisfied under the uniform random assignment?

(b) (Bonus points) What is the expected number of satisfied clauses under the uniform random assignment? Does this number depend on the number of variables? Does it depend on the number of clauses?

(c) (Bonus points) Let ϕ be an m-clause 3-CNF formula with distinct variables in each clause. Conclude from (b) that if $m < 8$ then there is at least one satisfying assignment for ϕ.

Hint: Recall that if a random variable X has $\mathbb{E}[X] = \mu$, there exists at least one realization in the domain of X whose value is at least μ.

(d) (Bonus points) Let $\mu(\phi)$ be the expected number of satisfied clauses you derived in (b). Show that the probability that a randomly chosen variable assignment satisfies at least $\mu(\phi)$ clauses is at least $\frac{1}{8m}$.